



Sydney Girls High School
2022
Trial Higher School Certificate
Examination

Mathematics Extension 1

**General
Instructions**

- Reading time – 10 minutes
- Working time – 2 hours
- Write using a black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 11-14, show relevant mathematical reasoning and/or calculations

Total marks:
70

Section I – 10 marks (pages 3-6)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 8-15)

- Attempt Questions 11-14
- Allow about 1 hour and 45 minutes for this section

Name:

.....

Teacher:

.....

THIS IS A TRIAL PAPER ONLY

It does not necessarily reflect the format
or the content of the 2022 HSC
Examination Paper in this subject.

Section 1

10 marks

Attempt Questions 1-10

Use the Multiple-choice answer sheet for questions 1-10

1) Given $f(x) = \sqrt{x} - 3$, what are the domain and range of $y = f^{-1}(x)$?

- A. $x \geq -3, y \geq 0$
- B. $x \geq -3, y \geq -3$
- C. $x \geq 0, y \geq 0$
- D. $x \geq 0, y \geq -3$

2) What is the derivative of $f(x) = \tan^{-1}\left(\frac{1}{x}\right)$?

- A. $\frac{-1}{1+x^2}$
- B. $\frac{-x^2}{1+x^2}$
- C. $\frac{1}{1+x^2}$
- D. $\frac{x^2}{1+x^2}$

3) Terry is one of 5 candidates in an election. If there are 106 voters what is the minimum number of votes she could receive and still win?

- A. 20
- B. 21
- C. 22
- D. 23

4) What is the solution set of the inequality $x(x^2 - 9) < x$?

- A. $(-\sqrt{8}, 0) \cup (\sqrt{8}, \infty)$
- B. $(-\infty, -\sqrt{10}) \cup (0, \sqrt{10})$
- C. $(-\infty, -\sqrt{8}) \cup (0, \sqrt{8})$
- D. $(-\sqrt{10}, 0) \cup (\sqrt{10}, \infty)$

5) A curve in the Cartesian plane is defined parametrically by the equations $x = t^3 + t$ and $y = t^4 + 2t^2$.

The equation of the tangent to the curve at $t = 1$ is:

- A. $y = 2x$
- B. $y = 8x$
- C. $y = 2x - 1$
- D. $y = 8x - 13$

6) What is the component of the force $\vec{F} = a\vec{i} + b\vec{j}$ where a and b are non-zero real constants, in the direction of the vector $\vec{w} = \vec{i} + \vec{j}$?

- A. $\left(\frac{a+b}{2}\right)\vec{w}$
- B. $\left(\frac{1}{a+b}\right)\vec{F}$
- C. $\left(\frac{a+b}{a^2+b^2}\right)\vec{F}$
- D. $\left(\frac{a+b}{\sqrt{2}}\right)\vec{w}$

- 7) Let R be the region between the graphs of $y = 1$ and $y = \sin x$ from $x = 0$ to $x = \frac{\pi}{2}$. The volume of the solid obtained by revolving R about the x -axis is given by:

A.
$$2\pi \int_0^{\frac{\pi}{2}} x \sin x \, dx$$

B.
$$\pi \int_0^{\frac{\pi}{2}} (1 - \sin x)^2 \, dx$$

C.
$$\pi \int_0^{\frac{\pi}{2}} \sin^2 x \, dx$$

D.
$$\pi \int_0^{\frac{\pi}{2}} (1 - \sin^2 x) \, dx$$

- 8) In how many different ways can three girls and three boys be seated on a bench for a photograph if the girls must all sit apart?

A. 72

B. 120

C. 144

D. 576

9) Given $\tan A - \tan B = \cot A - \cot B - 1 = m$, which of the following is the correct expression for $\tan(A - B)$ in terms of m .

A. $\frac{m(m+1)}{2m+1}$

B. $m(m + 1)$

C. $-\frac{m(m+1)}{2m+1}$

D. $-m(m + 1)$

10) Given that x and y are both prime numbers and integers larger than 1, how many terms of the binomial expansion $(\sqrt{x} + \sqrt[5]{y})^{301}$ will be rational?

A. 151

B. 60

C. 31

D. 30

Section II

60 marks

Attempt questions 11-14

Start each question on a NEW piece of paper

Question 11 (15 marks)

Use a new sheet of paper.

(a) Let $\overrightarrow{OA} = 3\mathbf{i} - 5\mathbf{j}$ and $\overrightarrow{OB} = 2\mathbf{i} + 6\mathbf{j}$.

(i) Find $|\overrightarrow{OA}|$. 1

(ii) Find $\overrightarrow{AB}$. 1

(iii) Find the size of $\angle AOB$, correct to the nearest degree. 2

(b) Find the value of the following integral, giving your answer in terms of π . 2

$$\int_0^{\frac{1}{3}} \frac{1}{\sqrt{4-9x^2}} dx$$

(c) A polynomial is given by $P(x) = x^3 + ax^2 - 6ax + b$. 3

Find the values of a and b given the following conditions.

- $x = 2$ is a root of the polynomial
- and
- the remainder is 2 when $P(x)$ is divided by $(x - 1)$

Question 11 continues on the following page

Question 11 (continued)

- (d) (i) Given $t = \tan\left(\frac{\theta}{2}\right)$, show that: 1

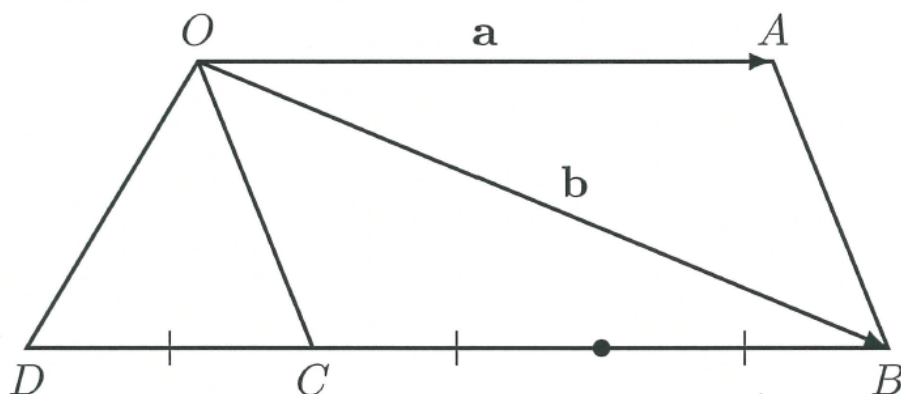
$$\sin\left(\frac{\theta}{2}\right) = \frac{t}{\sqrt{1+t^2}} .$$

- (ii) Hence solve the following equation for θ , where $\theta \in [0, 2\pi]$. 3

$$\cos \theta - \sin\left(\frac{\theta}{2}\right) = 0$$

Express your solutions in terms of π .

- (e) In the diagram below, $OABC$ is a parallelogram and D, C and B are collinear such that $DC:CB = 1:2$. Given $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OB} = \underline{b}$, find an expression for $\overrightarrow{AD}$ in terms of $\underline{a}$ and $\underline{b}$. 2



Question 12 (15 marks)**Use a new sheet of paper**

- (a) The rate at which a body warms in air is proportional to the difference between its temperature T and the constant temperature A of the surrounding air.

This rate can be expressed using the differential equation:

$$\frac{dT}{dt} = k(T - A)$$

where t is the time in minutes and k is a constant.

- (i) Show that $T = A + Pe^{kt}$, where P is a constant, is a solution of the differential equation. 1
- (ii) At the time a cheesecake is removed from a fridge, it is 2°C . 3
After 15 minutes, it has warmed to 10°C .
Given the air temperature around the cheesecake is 25°C , find the temperature of the cheesecake after a further 15 minutes.
Give your answer in degrees, correct to two decimal places.

(b)

- (i) Express $f(x) = \sqrt{3} \cos x - \sin x$ in the form $R \cos(x + \alpha)$, where $0 < \alpha < \frac{\pi}{2}$. 2
- (ii) Sketch $y = f(x)$ for $0 \leq x \leq 2\pi$, showing clearly the location of all stationary points and the end-points. 2
Your sketch should be at least one-third of a page.
- (iii) Hence, solve $\sqrt{3} \leq f(x) \leq 2$ for $0 \leq x \leq 2\pi$, expressing your solution using values in terms of π , where necessary. 1

Question 12 continues on the following page

Question 12 (continued)

- (c) Use the principle of mathematical induction to show that for all integers $n \geq 1$, 3

$$1^2 + 3^2 + 5^2 + \cdots + (2n - 1)^2 = \frac{4n^3 - n}{3} .$$

- (d) Muckle Flugga Lighthouse is located on a small island and the lighthouse is 3 located 3 km away from the main coastline. Assume the main coastline is a straight line. 3

Cooper is positioned at the point on the main coastline closest to the lighthouse.

If the light of the lighthouse revolves at 2 revolutions per minute, how fast is the beam of light moving along the coastline at a point 2 km away from Cooper?

Express your answer in terms of π .



Question 13 (15 marks)

Use a new sheet of paper

(a) (i) Sketch the function $y = f(x)$ where $f(x) = 9x - x^3$. 2

(ii) Hence, sketch $|y| = f(x - 3)$. 2

(b) Use the substitution $u = \tan x$ to find the following integral: 2

$$\int \frac{\operatorname{cosec} x}{\cos x} dx$$

(c) Find the coefficient of x^3 upon expansion of the following: 3

$$\left(1 + \frac{2}{x^2}\right)^2 (2 - x)^{10}$$

Question 13 continues on the following page

Question 13 (continued)

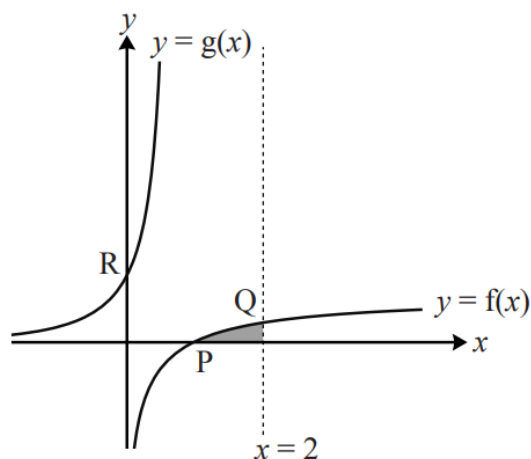
(d) The given diagram shows the curves

$y = f(x)$ and $y = g(x)$ where:

$$f(x) = \ln\left(\frac{2x}{1+x}\right), \quad x > 0$$

$$g(x) = \frac{e^x}{2 - e^x}, \quad x < \ln 2$$

The curve $y = f(x)$ crosses the x -axis at $P(1,0)$, and the line $x = 2$ at Q .



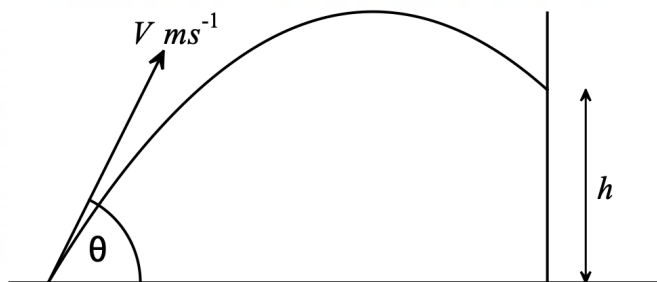
- | | | |
|-------|---|---|
| (i) | Determine the y -coordinate of the point Q . | 1 |
| (ii) | Find $f^{-1}(x)$, the inverse function of $f(x)$. | 1 |
| (iii) | Show, using the substitution $u = 2 - e^x$, that: | 2 |

$$\int_0^{\ln \frac{4}{3}} g(x) dx = \ln \frac{3}{2}$$

- | | | |
|------|--|---|
| (iv) | Hence, find the area of the shaded region in the diagram, giving your answer in the form $\ln \frac{a}{b}$ where a and b are integers. | 2 |
|------|--|---|

Question 14 (15marks)**Use a NEW sheet of paper**

- (a) A rocket is fired with a velocity $V \text{ ms}^{-1}$ at an angle θ to the horizontal from sea level and hits a cliff face as shown. Neglect air resistance and take $g = 10 \text{ m/s}^2$.



The position vector of the rocket, relative to the point from which it was fired is given by:

$$\underline{s}(t) = 20t\underline{i} + (20\sqrt{3}t - 5t^2)\underline{j}$$

where t is the time in seconds after the rocket was fired.

- (i) The base of the cliff is 120 m from the point at which the rocket was fired. 2

Find the value of h , correct to the nearest metre.

- (ii) Find the angle that the rocket makes with the cliff at the moment it strikes the cliff. 2

Give your answer correct to the nearest minute.

- (b) Consider the functions $y = -\frac{1}{2}\tan^{-1}(-2x)$ and $y = \cos^{-1}(2x)$.

- (i) Sketch the function $y = \cos^{-1}(2x)$. 1

- (ii) Find the coordinates of any points of intersection and provide all values correct to 3 significant figures. 3

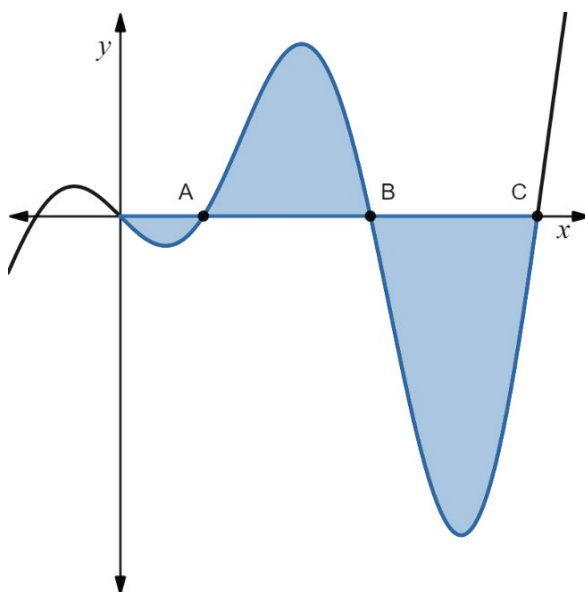
Question 14 continues on the following page

Question 14 (continued)

(c) The diagram below represents the graph of the function $y = \sin 5x - \sin 6x$.

(i) Determine the exact coordinates of the point C . 2

(ii) Hence, calculate the volume of the solid formed when the shaded region is rotated by 360° about the x -axis. 3
Give your answer correct to 2 decimal places.

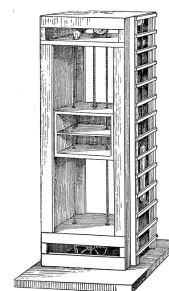


(d) The letters of the word DUMBWAITERS are arranged randomly.

Find the probability that:

(iii) the word **WEIRD** appears in the arrangement so that the letters are both in order and all together.

For example, the arrangement TUMBWEIRDAS.



1

(iv) the letters of the word **WEIRD** appear in the arrangement so that the letters are in order but not all together

For example, the arrangement UWTMEABIRDS.

1

End of paper

SGHS X1 Trial HSC 2022: SOLUTIONS

Q01 A For $f(x)$, the domain is $x \geq 0$ and the range is $y \geq -3$.
Hence, for $f^{-1}(x)$, the domain is $x \geq -3$ and the range is $y \geq 0$.

Q02 A

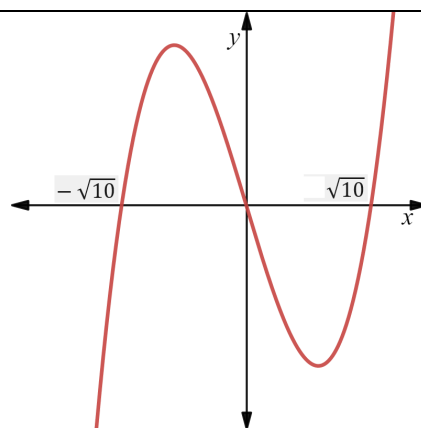
$$f'(x) = \frac{1}{1 + \left(\frac{1}{x}\right)^2} \times -x^{-2}$$

$$= \frac{1}{1 + \frac{1}{x^2}} \times -\frac{1}{x^2}$$

$$\therefore f'(x) = -\frac{1}{x^2 + 1}$$

Q03 C $\frac{106}{5} = 21.2$
 $\therefore$ The minimum number of votes Terry can receive and still win is 22.

Q04 B $x(x^2 - 10) < 0$
 $x(x - \sqrt{10})(x + \sqrt{10}) < 0$
Using the graph on the right:
 $x < -\sqrt{10}$ or $0 < x < \sqrt{10}$
i.e. $(-\infty, -\sqrt{10}) \cup (0, \sqrt{10})$



Q05 C

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= \frac{4t^3 + 4t}{3t^2 + 1}$$

when $t = 1, x = 2, y = 3$

$$m_T = \frac{4 + 4}{3 + 1}$$

$$m_T = 2$$

Eqn of tangent is $y - 3 = 2(x - 2)$
i.e. $y = 2x - 1$

Q06

A

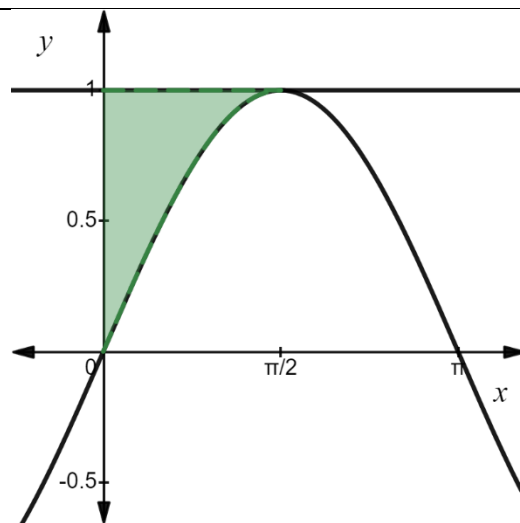
$$\begin{aligned} \text{proj}_{\tilde{w}} F &= \frac{\tilde{F} \cdot \tilde{w}}{\tilde{w} \cdot \tilde{w}} \tilde{w} \\ &= \frac{a \times 1 + b \times 1}{1 \times 1 + 1 \times 1} \tilde{w} \end{aligned}$$

$$\therefore \text{proj}_{\tilde{w}} F = \frac{a+b}{2} \tilde{w}$$

Q07

D

$$\begin{aligned} V &= \pi \int_0^{\frac{\pi}{2}} 1^2 dx - \pi \int_0^{\frac{\pi}{2}} (\sin x)^2 dx \\ \therefore V &= \pi \int_0^{\frac{\pi}{2}} (1 - \sin^2 x) dx \end{aligned}$$



Q08

C

No. of ways to order the three boys = 3!

_B_B_B_ No. of seating choices for the three girls = $4 \times 3 \times 2$

$\therefore$ Total no of ways for seating = $3! \times 24 = 144$

Q09

B

$$\tan A - \tan B = m$$

$$\cot A - \cot B - 1 = m$$

$$\frac{1}{\tan A} - \frac{1}{\tan B} = m + 1$$

$$\frac{\tan B - \tan A}{\tan A \tan B} = m + 1$$

$$\frac{-m}{\tan A \tan B} = m + 1$$

$$\tan A \tan B = \frac{-m}{m+1}$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$= \frac{m}{1 - \frac{m}{m+1}}$$

$$= \frac{m(m+1)}{m+1-m}$$

$$\therefore \tan(A - B) = m(m+1)$$

Q10

D

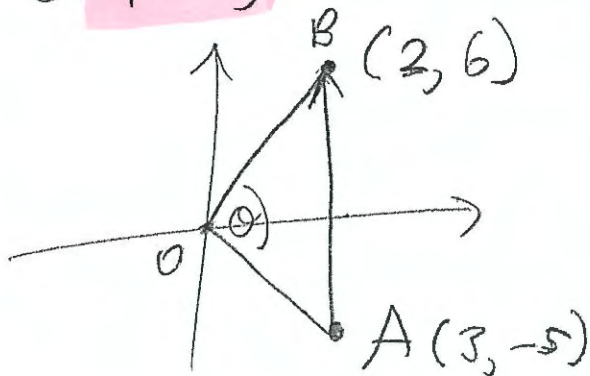
There will be 302 terms containing $\left((x)^{\frac{1}{2}}\right)^a \left((y)^{\frac{1}{5}}\right)^b$

where $a + b = 301$ and $0 \leq a, b \leq 301$.

The rational terms occur when a is a multiple of 2 **and** b is a multiple of 5. Since a will be even and $a + b$ is odd, b must be odd.

Hence, there will be 30 rational terms, when $b = 5, 15, 25, \dots, 295$.

Q 11 a)



$$(i) |\vec{OA}| = \sqrt{3^2 + 5^2} \\ = \sqrt{34} \text{ units}$$

$$(ii) \vec{AB} = \vec{AO} + \vec{OB} \\ = -\vec{OA} + \vec{OB} \\ = -(3\hat{i} - 5\hat{j}) + 2\hat{i} + 6\hat{j} \\ = -\hat{i} + 11\hat{j}$$

$$(iii) OA = \sqrt{34} \text{ units} \\ OB = \sqrt{2^2 + 6^2} \\ = \sqrt{40}$$

$$\text{Let } \angle AOB = \theta \\ \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$\text{RHS} = \sqrt{34} \times \sqrt{40} \times \cos \theta \\ = \sqrt{1360} \cos \theta$$

$$\text{LHS} = \vec{a} \cdot \vec{b} \\ = x_1 x_2 + y_1 y_2 \\ = 3 \times 2 + -5 \times 6 \\ = 6 - 30 \\ = -24$$

$$\cos \theta = \frac{-24}{\sqrt{1360}}$$

$$\theta \doteq 49^\circ$$

$$\angle AOB = 180^\circ - 49^\circ \\ \doteq 131^\circ$$

(to nearest deg.)

11(b)

$$\int_0^{\frac{1}{3}} \frac{1}{\sqrt{4-9x^2}} dx$$

$$= \frac{1}{3} \int_0^{\frac{1}{3}} \frac{3}{\sqrt{2^2 - (3x)^2}} dx$$

$$= \frac{1}{3} \left[\sin^{-1} \left(\frac{3x}{2} \right) \right]_0^{\frac{1}{3}}$$

$$= \frac{1}{3} \left(\sin^{-1} \left(\frac{1}{2} \right) - \sin^{-1}(0) \right)$$

$$= \frac{1}{3} \left(\frac{\pi}{6} - 0 \right)$$

$$= \frac{\pi}{18}$$

/2

11 c) $p(x) = x^3 + ax^2 - 6ax + b$

$$p(2) = 0$$

$$p(1) = 2$$

$$8 + 4a - 12a + b = 0$$

$$8a - b = 8 \dots\dots ①$$

$$1 + a - 6a + b = 2$$

$$5a - b = -1 \dots\dots ②$$

$$① - ②: 3a = 9$$
$$a = 3$$

Sub a in ②:

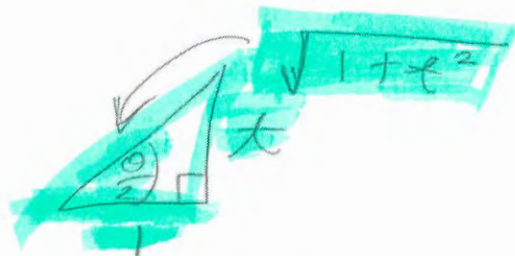
$$15 - b = -1$$

$$b = 16$$

$$\therefore a = 3, b = 16$$

3

11 d) (i) $x = \tan\left(\frac{\theta}{2}\right)$



1

$$\sin\left(\frac{\theta}{2}\right) = \frac{x}{\sqrt{1+x^2}}$$

(ii) $\cos \theta = \frac{1-x^2}{1+x^2}$

$$\cos \theta - \sin\left(\frac{\theta}{2}\right) = 0$$

$$\frac{1-x^2}{1+x^2} = \frac{x}{\sqrt{1+x^2}}$$

$$1-x^2 = \frac{x(1+x^2)}{(1+x^2)^{\frac{1}{2}}}$$

$$1-x^2 = x(1+x^2)^{\frac{1}{2}}$$

$$(1-x^2)^2 = x^2(1+x^2)$$

$$1-2x^2+x^4 = x^2+x^4$$

$$3x^2 = 1$$

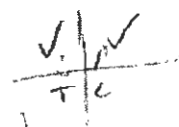
$$x^2 = \frac{1}{3}$$

$$x = \pm \frac{1}{\sqrt{3}}$$

$$0 \leq \theta \leq 2\pi$$

$$0 \leq \frac{\theta}{2} \leq \pi$$

$$\tan\left(\frac{\theta}{2}\right) = \pm \frac{1}{\sqrt{3}}$$



$$\frac{\theta}{2} = \frac{\pi}{6} \quad \pi - \frac{\pi}{6}$$

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

3

$$\text{11 e)} \quad \vec{AB} = \vec{AO} + \vec{OB} \\ = -\vec{a} + \vec{b}$$

$$\vec{AD} = \vec{AB} + \vec{BD} \\ = (\vec{b} - \vec{a}) + \frac{3}{2} \times -\vec{a} \\ = \vec{b} - \frac{5}{2} \vec{a}$$

1/2

OR

$$\vec{OD} = \vec{OB} + \vec{BD} \\ = \vec{b} + -\frac{3}{2} \vec{a}$$

$$\vec{AB} = \vec{AO} + \vec{OD} \\ = -\vec{a} + (\vec{b} - \frac{3}{2} \vec{a}) \\ = \vec{b} - \frac{5}{2} \vec{a}$$

Question 12

(a)

$$(i) \quad T = A + Pe^{kt} \quad (1)$$

$$\frac{dT}{dt} = kPe^{kt}$$

$$= k(T-A), \text{ since } Pe^{kt} = T-A \text{ from (1)}$$

$\therefore$ (1) satisfies the DE. ✓

Comment: Generally well done. There were a handful of students that were not successful with this question - it is recommended to learn the above setting out.

$$(ii) \quad A = 25$$

$$T = 25 + Pe^{kt}$$

$$t=0, T=2$$

$$2 = 25 + Pe^0$$

$$\therefore -23 = P \quad \checkmark$$

$$T = 25 - 23e^{kt}$$

$$t=15, T=10$$

$$10 = 25 - 23e^{15k}$$

$$-15 = -23e^{15k}$$

$$e^{15k} = \frac{15}{23}$$

$$15k = \ln \frac{15}{23}$$

$$\therefore k = \frac{1}{15} \ln \frac{15}{23} \quad \checkmark$$

$$\text{When } t=30,$$

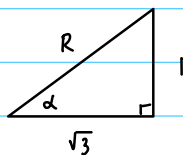
$$\begin{aligned} T &= 25 - 23e^{30k} \\ &= 15.22^\circ\text{C} \quad (2 \text{ dp}) \quad \checkmark \end{aligned}$$

(b)

(i) $\sqrt{3} \cos x - \sin x = R \cos(x + \alpha)$
 $= R \cos x \cos \alpha - R \sin x \sin \alpha$

$$R \cos \alpha = \sqrt{3}$$

$$R \sin \alpha = 1$$



$$R = \sqrt{(\sqrt{3})^2 + 1^2}$$

$$= 2$$

$$\tan \alpha = \frac{1}{\sqrt{3}}$$

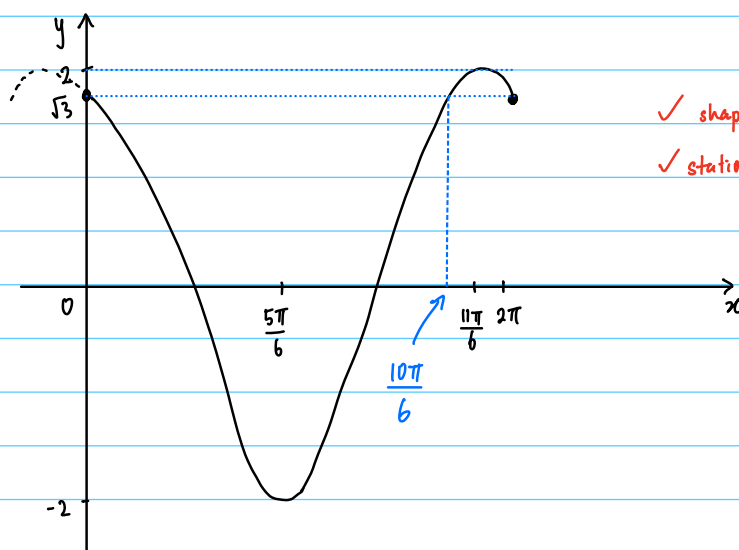
$$\alpha = \frac{\pi}{6}$$

✓✓

$$\therefore f(x) = 2 \cos\left(x + \frac{\pi}{6}\right)$$

Comment: Generally well done. Proper derivation of the values of R and α are needed to obtain full marks.

(ii)



✓ shape and position

✓ stationary and endpoints

Comment: Students who recognised that $f(x) = 2 \cos(x + \frac{\pi}{6})$ is simply a shift of $y = 2 \cos x$ by $\frac{\pi}{6}$ units to the left were generally the most successful.

Many students inefficiently derived the coordinates of the stationary points by solving trigonometric equations and/or by differentiating $f(x)$.

(iii) $x = 0, \frac{5\pi}{3} \leq x \leq 2\pi$ ✓

Comment: Many students left out $x=0$ as part of their solution.

(C) Prove true for $n=1$

$$\text{LHS} = 1^3 = 1$$

$$\text{RHS} = \frac{4(1)^3 - 1}{3} = 1$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore$ true for $n=1$ ✓

Assume true for $n=k$

ie. assume

$$1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 = \frac{4k^3 - k}{3}$$

Prove true for $n=k+1$

ie. RTP

$$1^2 + 3^2 + 5^2 + \dots + (2k+1)^2 = \frac{4(k+1)^3 - (k+1)}{3}$$

$$\text{LHS} = \frac{4k^3 - k}{3} + (2k+1)^2 \quad (\text{by assumption})$$

$$= \frac{4k^3 - k + 3(4k^2 + 4k + 1)}{3} \quad \checkmark$$

$$= \frac{4k^3 + 12k^2 + 12k + 4 - k - 1}{3}$$

$$= \frac{4(k^3 + 3k^2 + 3k + 1) - (k+1)}{3} \quad \checkmark$$

$$= \frac{4(k+1)^3 - (k+1)}{3}$$

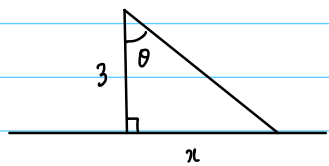
$$= \text{RHS}$$

$\therefore$ true for $n=k+1$

$\therefore$ true by induction for integers $n \geq 1$.

Comment: Students who knew that $(k+1)^3 = k^3 + 3k^2 + 3k + 1$ were generally the most successful.

(d)



✓ ($\theta, x, 3$ must ALL be labelled)

$$\frac{d\theta}{dt} = 2 \text{ revs/min}$$

$$= 4\pi \text{ rads/min}$$

$$\begin{aligned} \frac{dx}{dt} &= \frac{d\theta}{dt} \times \frac{dx}{d\theta} \\ &= 4\pi \times \frac{9+x^2}{3} \end{aligned}$$

$$\tan \theta = \frac{x}{3}$$

$$\theta = \tan^{-1} \frac{x}{3}$$

$$\frac{d\theta}{dx} = \frac{\frac{1}{3}}{1 + \frac{x^2}{9}} = \frac{3}{9+x^2}$$

When $x = 2$,

$$\frac{dx}{dt} = 4\pi \times \frac{9+2^2}{3}$$

$$= \frac{52\pi}{3} \text{ km/min} \quad \checkmark$$

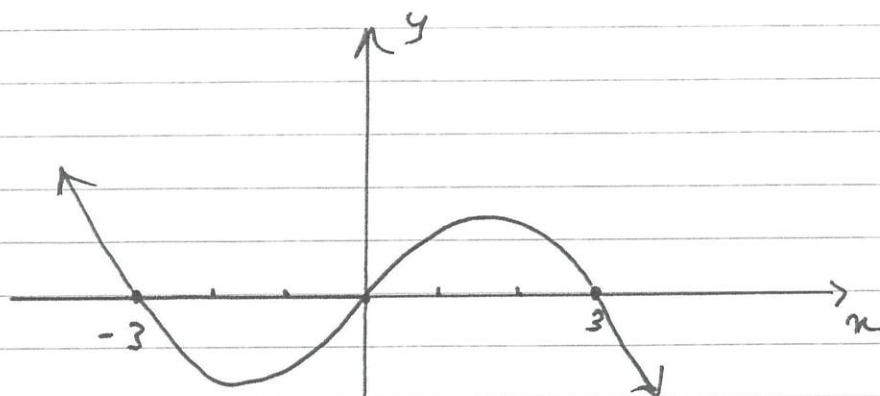
Comment: Only a handful of students were successful with this question. Students who were able to correctly depict the situation with a labelled diagram were generally the most successful.

Q13 Ext 1 2022 Trial

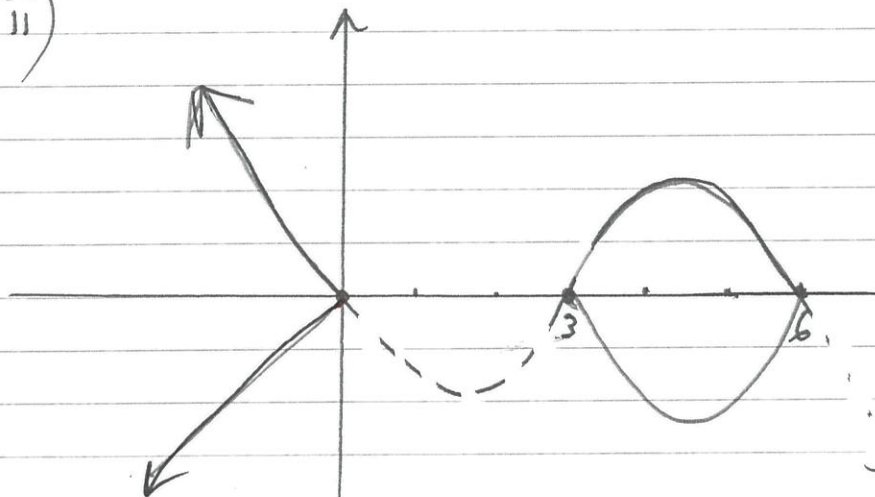
a) i) $y = 9x - x^3$

$$= x(9 - x^2)$$

$$= x(3 - x)(3 + x)$$



ii)



b) $u = \tan x$
 $\frac{du}{dx} = \sec^2 x$
 $du = \sec^2 x dx$

This question was done poorly. Many students didn't know how to do it.

$$\int \frac{1}{\sin x \cos x} \times \frac{\cos x}{\cos x} dx$$

$$= \int \frac{\cos x}{\sin x \cos^2 x} dx = \int \frac{\cos x \sec^2 x}{\sin x} dx$$

$$= \int \cot u \sec^2 u \, du$$

$$= \int \frac{1}{\tan u} \sec^2 u \, du$$

$$= \int \frac{1}{u} \, du$$

$$= \ln |u| + C$$

$$= \ln |\tan u| + C$$

✓✓

$$c) \left(1 + \frac{4}{x^2} + \frac{4}{x^4}\right) \left({}^{10}C_7 2^7 (-x)^3 + {}^{10}C_5 2^5 (-x)^5 + {}^{10}C_3 2^3 (-x)^7 \right)$$

$$= {}^{10}C_7 2^7 (-x)^3 + \frac{4}{x^2} \left({}^{10}C_5 2^5 (-x)^5 \right) + \frac{4}{x^4} \left({}^{10}C_3 2^3 (-x)^7 \right)$$

$$= -15360x^3 - 32256x^3 - 3840x^3$$

$$= -51456x^3$$

coefficient -51456

✓✓✓

This question was

very poorly done.

students didn't have clear

setting out.

$$d) i) y = \ln\left(\frac{2x}{1+x}\right)$$

$$x=2$$

$$y = \ln\left(\frac{4}{3}\right)$$

$$P\left(2, \ln\frac{4}{3}\right)$$

$$ii) y = \ln\left(\frac{2x}{1+x}\right)$$

$$x = \ln\left(\frac{2y}{1+y}\right)$$

$$e^x = \frac{2y}{1+y}$$

$$e^x + ye^x = 2y$$

$$e^x = 2y - ye^x$$

$$e^x = y(2 - e^x)$$

$$y = \frac{e^x}{2 - e^x}$$

$$iii) u = 2 - e^x$$

$$\frac{du}{dx} = -e^x$$

$$\int_0^{\ln\frac{4}{3}} \frac{e^x}{2 - e^x} dx$$

$$- \int_0^{\ln\frac{4}{3}} \frac{-e^x}{2 - e^x} dx$$

$$= \int \frac{du}{u}$$

$$u = 2 - e^x$$

$$x=0 \rightarrow u = 2 - 1$$

$$u = 1$$

$$x = \ln\frac{4}{3} \rightarrow u = 2 - e^{\ln\frac{4}{3}}$$

$$= 2 - \frac{4}{3}$$

$$= \frac{2}{3}$$

$$= \int_1^{\frac{2}{3}} \frac{du}{u}$$

$$= \left[\ln|u| \right]_1^{\frac{2}{3}}$$

$$= - \left[\ln\left(\frac{2}{3}\right) - \ln(1) \right]$$

$$= - \ln\left(\frac{2}{3}\right)$$

$$= \ln\left(\frac{2}{3}\right)^{-1}$$

$$= \ln\frac{3}{2}$$

iv)

$$A = \left(\ln \frac{4}{3}\right) \times 2 - \int_0^{\ln \frac{4}{3}} \frac{e^y}{2 - e^y} dy$$

$$= 2 \ln \frac{4}{3} - \left[\ln \frac{3}{2} \right]$$

$$= \ln \frac{16}{9} - \ln \frac{3}{2}$$

$$= \ln \left(\frac{\frac{16}{9}}{\frac{3}{2}} \right)$$

$$= \ln \left(\frac{32}{27} \right) u^2$$

This question was
hard for many
students

Q14

a) i) $s(t) = 20t \hat{i} + (20\sqrt{3}t - 5t^2) \hat{j}$

$$x = 20t \quad (1)$$

$$y = 20\sqrt{3}t - 5t^2 \quad (2)$$

from (1) $t = \frac{x}{20}$ subs into (2)

$$y = 20\sqrt{3} \times \frac{x}{20} - \frac{5 \times x^2}{400} \checkmark$$

$$y = \sqrt{3}x - \frac{x^2}{80}$$

When $x = 120$

$$y = \sqrt{3} \times 120 - \frac{120^2}{80} = \boxed{28 \text{ m}} \checkmark$$

ii) When it hits the cliff $x = 120 \text{ m}$

$$t = \frac{120}{20} = 6 \text{ sec}$$

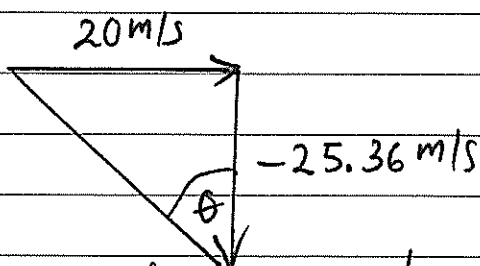
$$x = 20t$$

$$y = 20\sqrt{3}t - 5t^2$$

$$\dot{x} = 20 \text{ m/s}$$

$$\dot{y} = 20\sqrt{3} - 10t \checkmark$$

$$\text{At } t = 6 \therefore \dot{y} = 20\sqrt{3} - 10 \times 6$$
$$\dot{y} = -25.36 \text{ m/s}$$



$$\tan \theta = \left| \frac{20}{-25.36} \right|$$

$$\theta = \boxed{38^\circ 16'} \checkmark$$

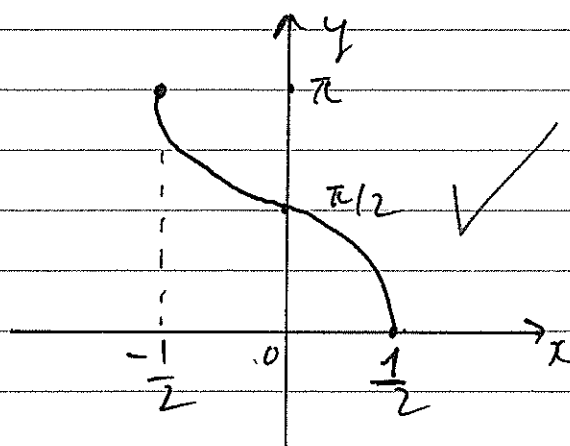
Note: To find the angle, the horizontal vel and vertical vel must be calculated (Not horizontal and vertical distances)

Q14

b) i) $y = \cos^{-1} 2x$

D: $-1 \leq 2x \leq 1$

R: $0 \leq y \leq \pi$



ii) solve $y = -\frac{1}{2} \tan^{-1}(-2x)$, $y = \cos^{-1}(2x)$

$\cos^{-1}(2x) = -\frac{1}{2} \tan^{-1}(-2x)$

$\cos^{-1}(2x) = \frac{1}{2} \tan^{-1}(2x)$ ✓

$2 \cos^{-1}(2x) = \tan^{-1}(2x)$

Insert tan for both sides

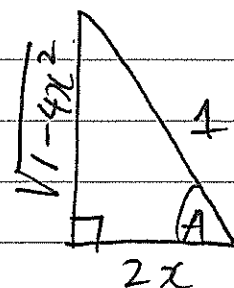
$\tan 2[\cos^{-1}(2x)] = \tan(\tan^{-1}(2x))$ (*)

$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$

let $A = \cos^{-1} 2x$

$\cos A = 2x$

$\tan A = \frac{\sqrt{1-4x^2}}{2x}$



from (*)

$\frac{2x \sqrt{1-4x^2}}{2x} = 2x$ ✓

$1 - \frac{(1-4x^2)}{4x^2}$

$$\frac{4x\sqrt{1-4x^2}}{8x^2-1} = 2x$$

$$2\sqrt{1-4x^2} = 8x^2 - 1$$

$$4(1-4x^2) = 64x^4 - 16x^2 + 1$$

$$4 - 16x^2 = 64x^4 - 16x^2 + 1$$

$$64x^4 = 3$$

$$x = \pm \sqrt[4]{\frac{3}{64}}$$

$$x = 0.465 \rightarrow y = 0.375$$

At $x = -0.465$ (two fns do not intersect)

Point of intersection: $(0.465, 0.375)$ ✓

c)

i) ~~$y = 5x$~~

At C, $y = 0$ $y = \sin 5x - \sin 6x$

$$\sin 5x - \sin 6x = 0$$

Use: $\sin(A-B) - \sin(A+B) = -2\cos A \cdot \sin B$

$$\sin 5x - \sin 6x = -2\cos \frac{11x}{2} \cdot \sin \frac{x}{2} \checkmark$$

OR $-2\cos \frac{11x}{2} \cdot \sin \frac{x}{2} = 0$

$$\begin{cases} \cos \frac{11x}{2} = 0 \\ \sin \frac{x}{2} = 0 \end{cases} \therefore \begin{cases} x = \frac{\pi}{11}, \frac{3\pi}{11}, \frac{5\pi}{11} \dots \\ x = 0, 2\pi, 4\pi \dots \end{cases}$$

Arrange the solⁿ in order

$$x = 0, \frac{\pi}{11}, \frac{3\pi}{11}, \frac{5\pi}{11} \dots$$

At point C, the 4th solution

$$\therefore x = \frac{5\pi}{11} \checkmark \text{ at } C$$

$$C \left(\frac{5\pi}{11}, 0 \right)$$

$$\text{ii)} \quad V = \pi \int_0^{\frac{5\pi}{11}} (\sin 5x - \sin 6x)^2 dx$$

$$V = \pi \int_0^{5\pi/11} (\sin^2 5x + \sin^2 6x - 2 \sin 5x \cdot \sin 6x) dx \checkmark$$

Note: $\sin^2 A = \frac{1}{2} (1 - \cos 2A)$

$$\sin^2 5x = \frac{1}{2} (1 - \cos 10x)$$

$$\sin^2 6x = \frac{1}{2} (1 - \cos 12x)$$

$$- 2 \sin 5x \cdot \sin 6x = -2 \left[\frac{1}{2} (\cos(-x) - \cos 11x) \right]$$

$$V = \pi \int_0^{5\pi/11} \left(\frac{1}{2} - \frac{\cos(10x)}{2} + \frac{1}{2} - \frac{\cos(12x)}{2} - \cos x + \cos 11x \right) dx$$

$$V = \pi \left[x - \frac{\sin(10x)}{20} - \frac{\sin(12x)}{24} - \sin x + \frac{\sin 11x}{11} \right]_0^{5\pi/11} \checkmark$$

Q14

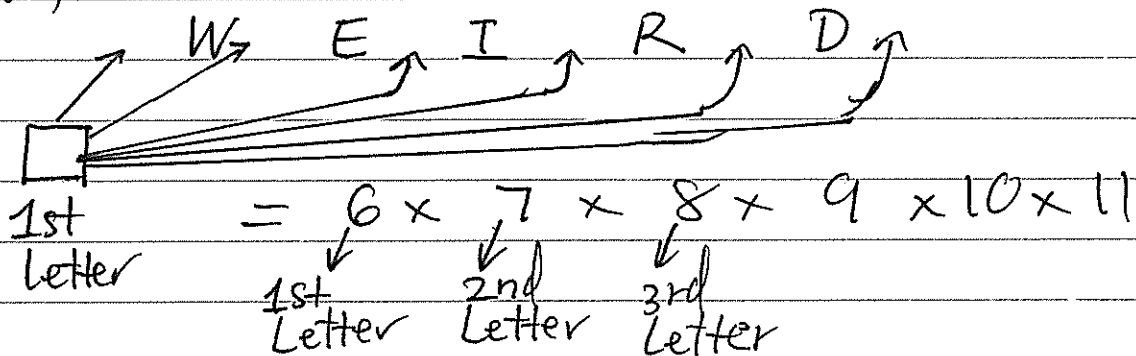
$$V = \pi \left[\frac{5\pi}{11} - \frac{1}{20} \sin\left(\frac{50\pi}{11}\right) - \frac{1}{24} \sin\left(\frac{60\pi}{11}\right) - \frac{\sin 5\pi}{11} + \frac{1}{11} \sin 5\pi \right]$$

$$V = 1.35 \text{ cubic units}$$

d) i) $\square \square \text{WEIRD} \square \square \square \square$

$$= \frac{7!}{11!} = \frac{1}{7920}$$

ii)



$P(\text{WEIRD IN ORDER But Not All Together})$

= All possible of WEIRD in order - WEIRD in order and together.

$$= \frac{6 \times 7 \times 8 \times 9 \times 10 \times 11}{11!} - \frac{7!}{11!}$$

$$= \frac{13}{1584}$$